Digital Signal Processing

Zhou Hang

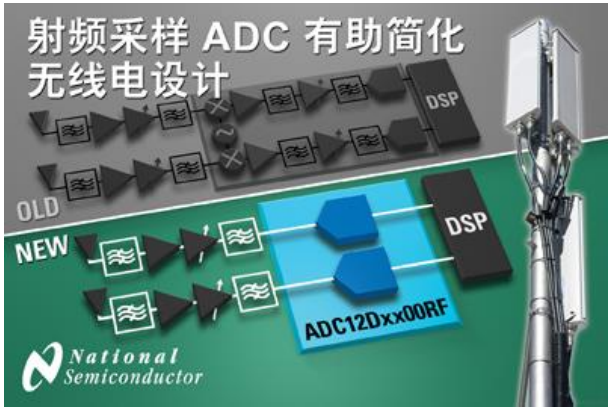
The Application of FFT in Wireless Communication

Lancaster University College at Beijing Jiaotong University

The rapid development of DSP technology ⇒ All digital receiver (software defined radio technology)



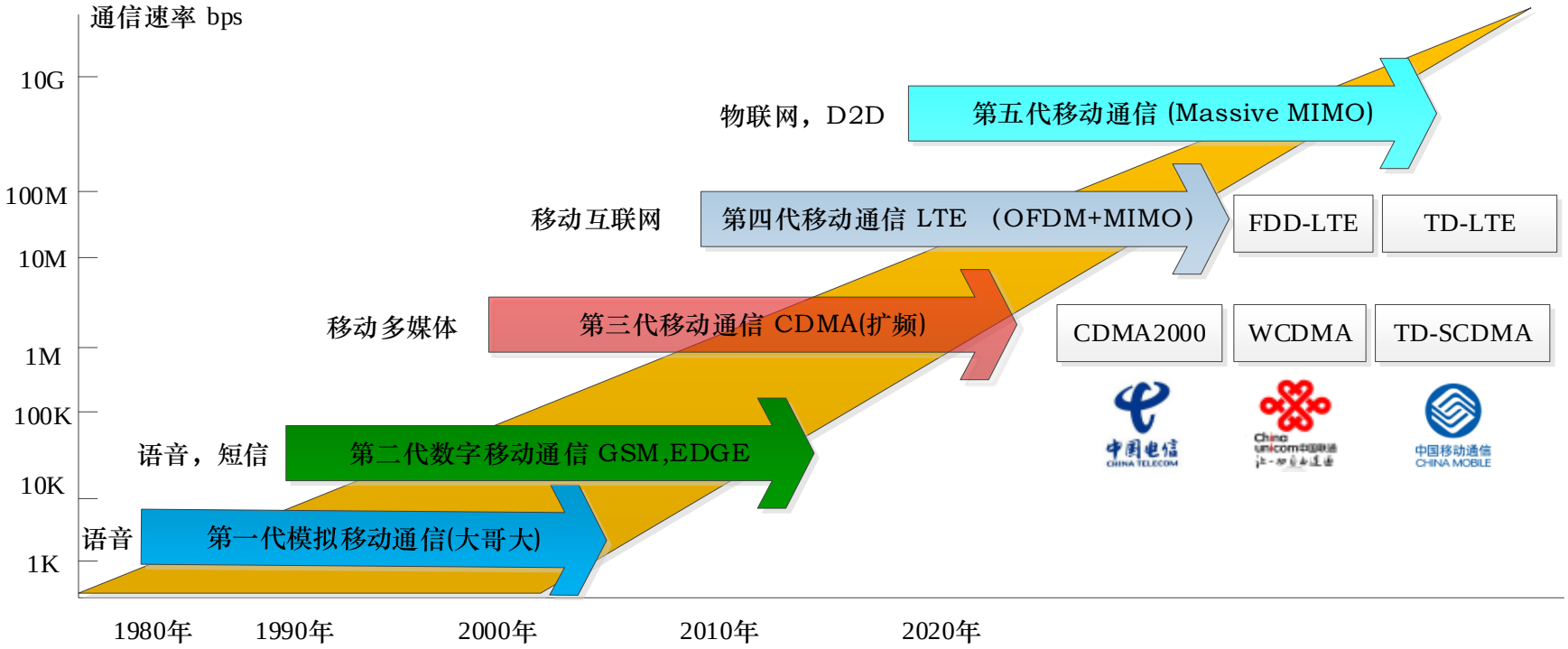
Software Defined Radio
SDR



1. An open, standardized, and modular universal hardware platform
2. Broadband A/D and D/A converters should be placed as close to the antenna as possible
3. Modulation and demodulation, filtering, encoding and decoding, encryption, etc. are all completed by software on programmable chips such as FPGA and DSP
4. Software download to upgrade functionality

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OFDM (Orthogonal Frequency-Division Multiplexing)

- **In the 1970s, Weistein, Ebert, and others developed a multi carrier transmission system called Orthogonal Frequency Division Multiplexing (OFDM) using DFT (FFT).**
- **The application of Discrete Fourier Transform (DFT) and its inverse transform (IDFT) methods in OFDM solves the problem of generating multiple orthogonal subcarriers and recovering the original signal from the subcarriers.**
- **Advantages: Combat echo interference caused by multipath channels**



Long Term Evolution



Wireless Fidelity

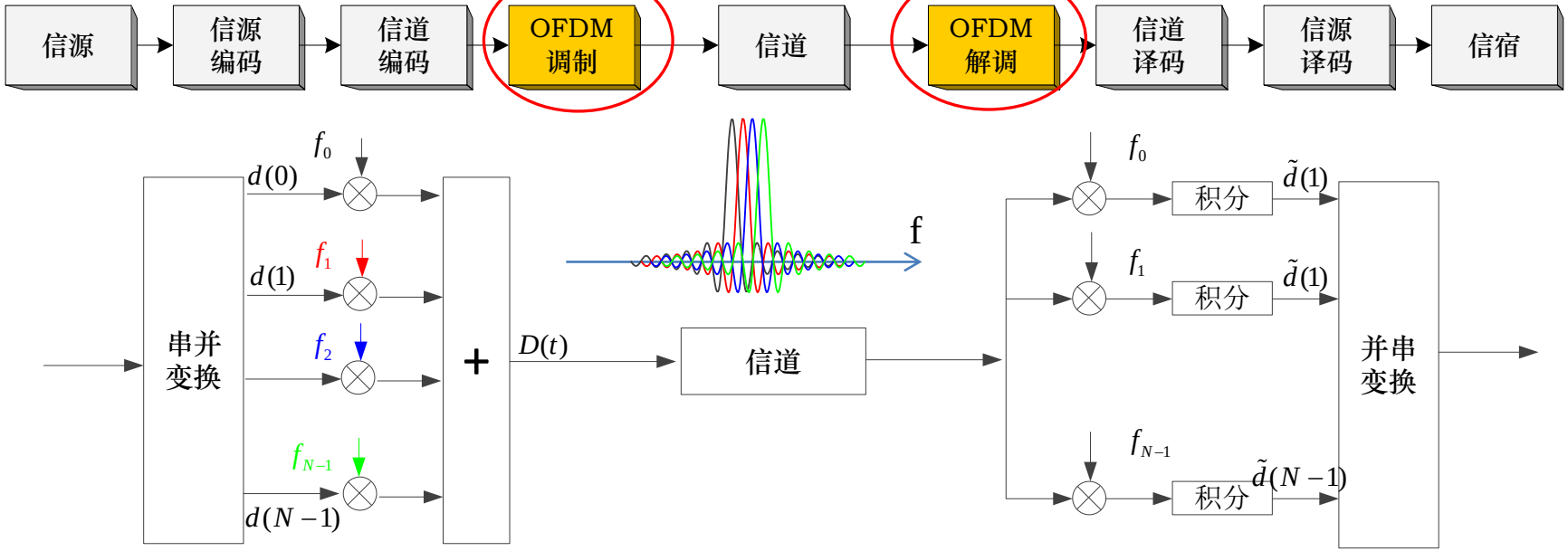


Asymmetric Digital Subscriber Line

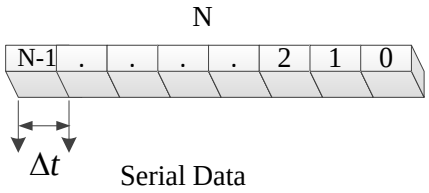


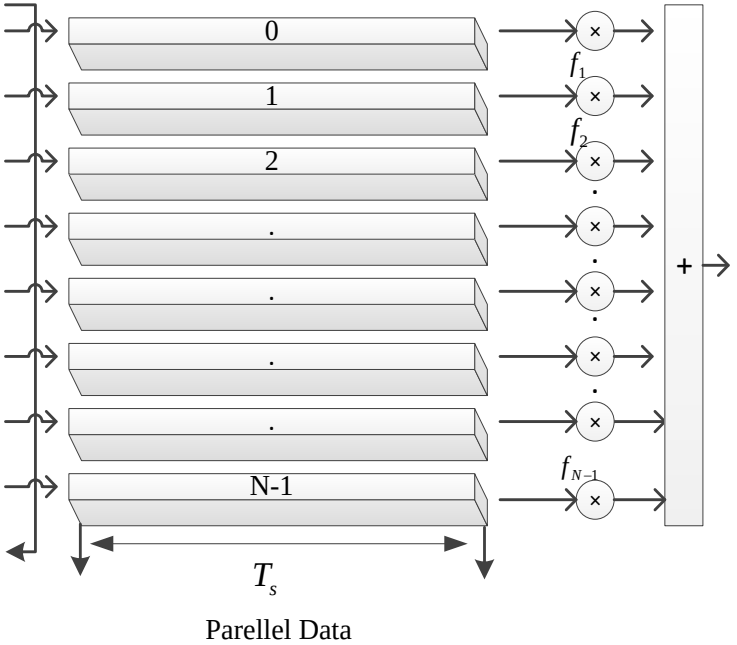
OFDM (Orthogonal Frequency-Division Multiplexing)

- Transmitting end: A data stream is divided into multiple parallel sub data streams and transmitted in parallel using multiple orthogonal subcarriers
- Receiver: Mixing integration separation of each sub channel (sub carrier)



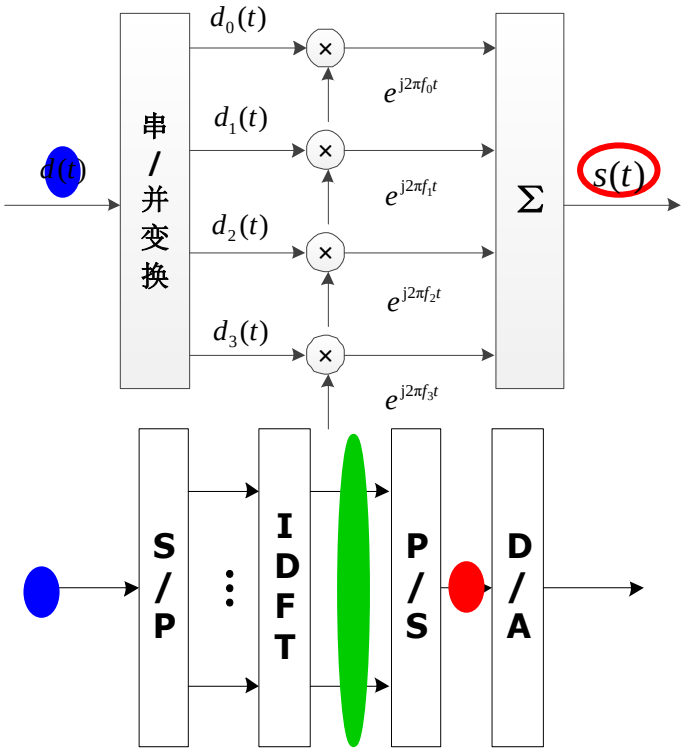
OFDM (Orthogonal Frequency-Division Multiplexing)





- Parallel symbol period: $T_s = N \cdot \Delta t$
- Subcarrier spacing: $\Delta f = 1/T_s$
- Subcarrier bandwidth: $2\Delta f$

OFDM transmitter



analog: $$s(t)=\sum_{n=0}^{N-1} d_n \cdot e^{j2\pi f_n t} \cdot rect(t)$$

$$s_k \overset{def}{=} s(k\cdot\Delta t) \qquad \Delta t = \frac{T_S}{N} \qquad k=0,1,\cdots,N-1$$

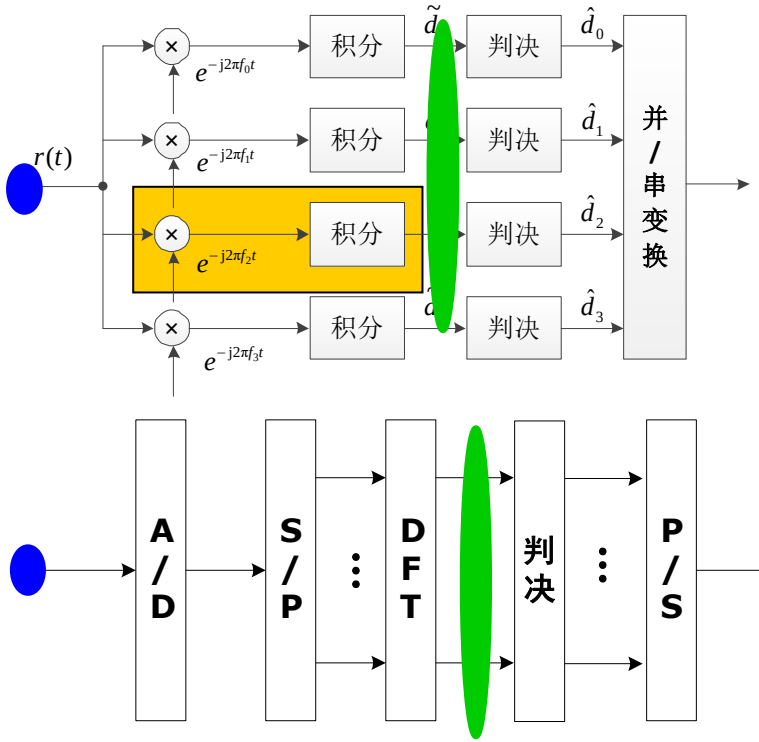
digital: $$s_k = \sum_{n=0}^{N-1} d_n \cdot e^{j2\pi f_n k\Delta t} \cdot rect(k\Delta t)$$

$$= \sum_{n=0}^{N-1} d_n \cdot e^{j2\pi \frac{n}{T_s} k\Delta t} \cdot 1$$

$$= \sum_{n=0}^{N-1} d_n \cdot e^{j2\pi \frac{n}{T_s} k \frac{T_s}{N}} = N\left(\frac{1}{N}\sum_{n=0}^{N-1} d_n \cdot e^{j2\pi \frac{k}{N} n}\right)$$

$$\mathbf{s} = \mathrm{IDFT}_N[\mathbf{d}] \qquad \begin{cases} \mathbf{d} = [d_0, d_1, \cdots, d_{N-1}] \\ \mathbf{s} = [s_0, s_1, \cdots, s_{N-1}] \end{cases}$$

OFDM receiver



analog: $\tilde{d}_m = \int_{t=0}^{T_s} r(t)\cdot e^{-j2\pi f_m t}\,dt$

$r_k \overset{def}{=} r(k\cdot\Delta t) \qquad \Delta t = \frac{T_S}{N} \qquad k = 0,1,\cdots,N-1$

digital: $\approx \sum_{k=0}^{N-1} r(k\cdot\Delta t)\cdot e^{-j2\pi f_m k\Delta t}\cdot\Delta t$

$= \sum_{k=0}^{N-1} r_k \cdot e^{-j2\pi\frac{m}{T_s}k\frac{T_S}{N}}\cdot\frac{T_S}{N}$

$= \frac{T_S}{N}\left(\sum_{k=0}^{N-1} r_k \cdot e^{-j2\pi\frac{m}{N}k}\right)$

$\mathbf{d} = \mathrm{DFT}_N[\mathbf{r}] \qquad \begin{cases}\mathbf{r} = [r_0, r_1, \cdots, r_{N-1}] \\ \tilde{\mathbf{d}} = [\tilde{d}_0, \tilde{d}_1, \cdots, \tilde{d}_{N-1}]\end{cases}$

FFT point count in current wireless communication standards



bandwidth	points
20(18) MHz	2048(1200)
15(13.5) MHz	1024(900)
10 (9)MHz	1024(600)
5 (4.5)MHz	512(300)



bandwidth	points
802.11a(20MHz)	64
802.11n(40MHz)	128
802.11ac(80MHz)	256
802.11ac(160MHz)	512



bandwidth	points
ADSL2(1MHz)	256
ADSL2+(2MHz)	512



bandwidth	points
192MHz	4096/8192
96MHz	2048/4096

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.